H1: Neutrino Oscillation Formula – Practical Units & MINOS

Griffiths Ch. 11 – From Eq. 11.3 to Eq. 11.9 in "Hybrid Natural Units"

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Problem Statement (H1, reworded)

Part (a). Begin with Griffiths Eq. 11.3 (the initial condition that the neutrino is born as a pure ν_e) and follow the derivation through to Griffiths Eq. 11.9 – the neutrino oscillation probability written in terms of the distance z traveled. Then *convert the units* of the argument of the second sine. Adopt this hybrid convention:

- $\Delta m^2 c^4$ measured in eV^2,
- L (the Griffiths "z") measured in km,
- E measured in GeV,
- $\hbar c$ kept in standard particle-physics natural units.

Show explicitly that: (i) all units cancel inside the argument of the second sine, as they must for it to be dimensionless, and (ii) a numerical factor of 1.27 appears inside that argument.

Part (b). Use the formula derived in (a), but on the right-hand side substitute Δm^2_{23} for Δm^2_{12} (i.e. pretend the two-state model now describes states 2 and 3 instead of 1 and 2). Take:

- $\sin^2(2\theta_{23}) = 1$ (maximal mixing, the measured value),
- Δm^2_{23} at the "Normal" Griffiths Fig. 11.5 value,
- $L = 730\,\text{km}$ (Fermilab → Soudan, MINOS baseline),
- $E_\nu = 3\,\text{GeV}$ (treat as a constant beam energy).

Compute: the fraction of muon neutrinos predicted to have oscillated (into ν_τ) at the MINOS far detector.

Deliverables

(a1) Briefly retrace the derivation Griffiths Eq. 11.3 → 11.9 so the starting point is unambiguous.
(a2) Insert numerical unit factors for $\Delta m^2 c^4$ [eV^2], L [km], E [GeV], and $\hbar c$.
(a3) Verify dimensional cancellation explicitly.
(a4) Show $\dfrac{10^3}{4\,(\hbar c \text{ in eV·m})\,10^9} = 1.2668\ldots \approx 1.27$.
(b1) State the value of $\Delta m^2_{23}{}^{\text{normal}}$ from Griffiths Fig. 11.5.
(b2) Plug numbers into the practical formula.
(b3) Compute $P(\nu_\mu \to \nu_\tau)$ – the fraction of ν_μ that have oscillated.
(b4) Comment briefly on what this implies physically about MINOS.

1 Background: Why This Formula Exists

Before any algebra, here's the physics in two paragraphs.

The flavor states ν_e, ν_μ, ν_τ that emerge from charged-current weak interactions (e.g. from $\pi^+ \to \mu^+ \nu_\mu$) are *not* eigenstates of the free Hamiltonian. The Hamiltonian eigenstates are the *mass eigenstates* ν_1, ν_2, ν_3 with definite masses m_1, m_2, m_3. Each ν_i propagates with its own definite phase $e^{-iE_i t/\hbar}$. If a neutrino is created in a flavor superposition (which is what ν_μ is), the relative phases between mass components evolve, and the projection onto a different flavor changes with distance traveled. That phase-driven flavor swap is what we call a *neutrino oscillation*.

For two flavors – a pedagogical simplification that does describe atmospheric and MINOS data well – the rotation between flavor and mass bases is parameterized by a single mixing angle θ, and the oscillation length is set by the energy splitting of the two mass eigenstates, which is $\Delta E \approx (\Delta m^2)c^4/(2E)$ for ultrarelativistic neutrinos.

Definition: Two-Flavor Mixing (Griffiths' Convention)

Griffiths writes Eq. 11.1 as

$$\nu_1 = -\sin\theta\,\nu_e + \cos\theta\,\nu_\mu, \qquad \nu_2 = \cos\theta\,\nu_e + \sin\theta\,\nu_\mu,$$

or in matrix form,

$$\begin{pmatrix}\nu_1\\ \nu_2\end{pmatrix} = \begin{pmatrix}-\sin\theta & \cos\theta\\ \cos\theta & \sin\theta\end{pmatrix}\begin{pmatrix}\nu_e\\ \nu_\mu\end{pmatrix}.$$

The cosine/sine choice automatically keeps the matrix orthogonal (probability-conserving). *Heads up:* this is slightly different from the convention used by H&M and Thomson, where the matrix is the standard rotation $(\cos\theta, \sin\theta; -\sin\theta, \cos\theta)$ – those texts thus end up with ν_1 aligned with ν_e at small θ, while Griffiths' ν_1 aligns with ν_μ. The final probability is identical; only the intermediate amplitudes differ in sign. Stick with one convention through any given problem.

2 Part (a): Derivation & Unit Conversion

2.1 Starting point: Griffiths Eq. 11.3

The problem says "start with Eq. 11.3." That equation is the *initial condition* – a neutrino is born at $t = 0$ as a pure ν_e:

$$\nu_e(0) = 1, \quad \nu_\mu(0) = 0 \iff \nu_1(0) = -\sin\theta, \quad \nu_2(0) = \cos\theta. \tag{Griffiths 11.3}$$

The right-hand equalities follow by inverting the mixing matrix above.

2.2 Time evolution of mass eigenstates

Because ν_1 and ν_2 *are* energy eigenstates, they pick up only phase factors:

$$\nu_1(t) = -\sin\theta\, e^{-iE_1 t/\hbar}, \qquad \nu_2(t) = \cos\theta\, e^{-iE_2 t/\hbar}. \tag{Griffiths 11.4}$$

2.3 Project back onto the ν_μ flavor

We want $\nu_\mu(t)$. Since Eq. 11.1 gives ν_1, ν_2 in terms of ν_e, ν_μ, we have to *invert* it. Write Eq. 11.1 in matrix form:

$$\begin{pmatrix}\nu_1\\ \nu_2\end{pmatrix} = M\begin{pmatrix}\nu_e\\ \nu_\mu\end{pmatrix}, \qquad M = \begin{pmatrix}-\sin\theta & \cos\theta\\ \cos\theta & \sin\theta\end{pmatrix}.$$

This M has $\det M = -\sin^2\theta - \cos^2\theta = -1$ and – conveniently – is *its own inverse* ($M^2 = I$, easy to check by direct multiplication). So inverting gives just the same matrix back:

$$\begin{pmatrix}\nu_e\\ \nu_\mu\end{pmatrix} = M\begin{pmatrix}\nu_1\\ \nu_2\end{pmatrix} \implies \boxed{\nu_\mu \;=\; \cos\theta\,\nu_1 \;+\; \sin\theta\,\nu_2.}$$

Sanity check the inversion

Plugging Eq. 11.3 in: $\nu_\mu(0) = \cos\theta\cdot(-\sin\theta) + \sin\theta\cdot\cos\theta = 0$ ✓, and $\nu_e(0) = -\sin\theta\cdot(-\sin\theta) + \cos\theta\cdot\cos\theta = \sin^2\theta + \cos^2\theta = 1$ ✓.

Now substitute the time-evolved amplitudes from Eq. 11.4:

$$\begin{aligned}\nu_\mu(t) &= \cos\theta\cdot\nu_1(t) \;+\; \sin\theta\cdot\nu_2(t)\\ &= \cos\theta\cdot\left(-\sin\theta\, e^{-iE_1t/\hbar}\right) \;+\; \sin\theta\cdot\left(\cos\theta\, e^{-iE_2t/\hbar}\right)\\ &= -\sin\theta\cos\theta\, e^{-iE_1t/\hbar} \;+\; \sin\theta\cos\theta\, e^{-iE_2t/\hbar}.\end{aligned}$$

Where the $\sin\theta\cos\theta$ cross terms come from. Each term in $\nu_\mu(t)$ is a product (coefficient from ν_μ formula) $\times$ (initial value $\nu_i(0)$):

	coefficient	$\nu_i(0)$	product
Term 1 ($e^{-iE_1t/\hbar}$)	$\cos\theta$	$-\sin\theta$	$-\sin\theta\cos\theta$
Term 2 ($e^{-iE_2t/\hbar}$)	$\sin\theta$	$\cos\theta$	$+\sin\theta\cos\theta$

Same magnitude $\sin\theta\cos\theta$, opposite signs. Factoring out:

$$\nu_\mu(t) = \sin\theta\cos\theta\left[e^{-iE_2t/\hbar} - e^{-iE_1t/\hbar}\right]. \qquad \text{(Griffiths 11.5)}$$

Common pitfall: where the $\sin^2\theta$ and $\cos^2\theta$ terms would come from

If instead you used $\nu_\mu = -\sin\theta\,\nu_1 + \cos\theta\,\nu_2$ – which is the inverse one would write under the standard H&M/Thomson convention but *not* Griffiths' – combined with Griffiths' initial conditions $\nu_1(0) = -\sin\theta$, $\nu_2(0) = \cos\theta$, you would get

$$\nu_\mu(t) \stackrel{?}{=} -\sin\theta\cdot(-\sin\theta)\, e^{-iE_1t/\hbar} + \cos\theta\cdot\cos\theta\, e^{-iE_2t/\hbar} = \sin^2\theta\, e^{-iE_1t/\hbar} + \cos^2\theta\, e^{-iE_2t/\hbar}.$$

This is not Eq. 11.5 – it is the *survival* amplitude $\nu_\mu \to \nu_\mu$ (probability of staying ν_μ if the particle started as ν_μ). The two cases differ by a sign convention applied inconsistently. The fix: under Griffiths' Eq. 11.1, the inverse for ν_μ is $\boxed{\nu_\mu = \cos\theta\,\nu_1 + \sin\theta\,\nu_2}$.

The probability of finding the neutrino as a ν_μ is $|\nu_\mu(t)|^2$. Using $|e^{ia} - e^{ib}|^2 = 2 - 2\cos(a-b) = 4\sin^2\left(\frac{a-b}{2}\right)$:

$$P(\nu_e \to \nu_\mu) = \sin^2(2\theta)\,\sin^2\left[\frac{(E_2 - E_1)\,t}{2\hbar}\right]. \qquad \text{(Griffiths 11.6)}$$

We used the identity $4\sin^2\theta\cos^2\theta = \sin^2(2\theta)$ for the prefactor.

2.4 Use the relativistic energy expansion

For a neutrino with momentum p and rest mass m_i,

$$E_i = \sqrt{p^2c^2 + m_i^2c^4} \;\approx\; pc + \frac{m_i^2c^4}{2pc} \;\approx\; E + \frac{m_i^2c^4}{2E},$$

where in the last step we use $pc \approx E$ (ultra-relativistic limit). Therefore:

$$E_2 - E_1 \approx \frac{(m_2^2 - m_1^2)\,c^4}{2E} = \frac{\Delta m^2 c^4}{2E}. \qquad \text{(Griffiths 11.7)}$$

Subtlety Worth Pausing On

The "standard" derivation (which Griffiths follows) holds p fixed – not E. As Kayser pointed out, this is technically wrong but "harmlessly" so: a careful wave-packet treatment gives the same final answer. The reason is that what really matters is the *phase difference*, and at high energy the leading-order term is $\Delta m^2c^4/(2E)$ regardless of whether you fix p or E.

2.5 Eq. 11.8: probability vs. time

Plugging $E_2 - E_1$ into Eq. 11.6:

$$P(\nu_e \to \nu_\mu) = \sin^2(2\theta)\,\sin^2\left[\frac{\Delta m^2 c^4}{4\hbar E}\,t\right]. \qquad \text{(Griffiths 11.8)}$$

2.6 Eq. 11.9: probability vs. distance z

Since $v \approx c$ for ultra-relativistic neutrinos, $z \approx ct$, so $t = z/c$. Substituting:

$$\boxed{P(\nu_e \to \nu_\mu) = \sin^2(2\theta)\,\sin^2\left[\frac{\Delta m^2 c^4}{4\,\hbar c\,E}\,z\right]} \qquad \text{(Griffiths 11.9)}$$

The combination $\hbar c$ in the denominator – with units of energy $\times$ length – is what we'll exploit when converting to practical units.

Why the 1/4 Appears

The phase difference accumulated by the wavefunction is $(E_2 - E_1)t/\hbar$. The probability formula contains $\sin^2$[half phase], hence the factor of 2 in the denominator. The other factor of 2 comes from $E_2 - E_1 = \Delta m^2c^4/(2E)$. Multiplied: 1/4. So the 4 in Griffiths Eq. 11.9 is 2×2, not arbitrary.

2.7 Now: the unit conversion

We want the dimensionless argument

$$\Phi \equiv \frac{\Delta m^2 c^4 \cdot z}{4\,\hbar c\, E}$$

expressed in mixed practical units. Insert each quantity with explicit units:

- $\Delta m^2 c^4$: numerical value X, in eV^2. Substitute $\Delta m^2 c^4 = X \cdot \mathrm{eV}^2$.
- $z = L$: numerical value Y, in km. Substitute $L = Y \cdot 10^3\,\mathrm{m}$ (because $1\,\mathrm{km} = 10^3\,\mathrm{m}$).
- E: numerical value Z, in GeV. Substitute $E = Z \cdot 10^9\,\mathrm{eV}$.
- $\hbar c$: in particle-physics natural units, $\hbar c = 197.327\,\mathrm{MeV \cdot fm}$. Convert to eV·m:

$$\hbar c = 197.327 \times 10^6\,\mathrm{eV} \times 10^{-15}\,\mathrm{m} = 1.97327 \times 10^{-7}\,\mathrm{eV \cdot m}.$$

Why $\hbar c$ in eV·m is the Right Choice

We're working with L in meters (after converting from km) and E in eV (after converting from GeV). To make the units inside $\hbar c \cdot E$ match the dimension energy^2×length, $\hbar c$ must be in eV·m. The conversion factor is straightforward: $197.327\,\mathrm{MeV} \times 10^6 = 1.973\,27 \times 10^8\,\mathrm{eV}$ per "MeV" part, times $10^{-15}\,\mathrm{m}$ per fm. Result: $\boxed{\hbar c = 1.973\,27 \times 10^{-7}\,\mathrm{eV \cdot m}}$.

Plugging everything in:

$$\begin{aligned}\Phi &= \frac{\Delta m^2 c^4 \cdot L}{4\,\hbar c\, E} \\ &= \frac{(X\,\mathrm{eV}^2) \cdot (Y \cdot 10^3\,\mathrm{m})}{4 \cdot (1.97327 \times 10^{-7}\,\mathrm{eV \cdot m}) \cdot (Z \cdot 10^9\,\mathrm{eV})}.\end{aligned} \tag{1}$$

2.8 Step 1: Verify the units cancel

Track only units in Eq. (1):

$$\frac{[\mathrm{eV}^2] \cdot [\mathrm{m}]}{[\mathrm{eV \cdot m}] \cdot [\mathrm{eV}]} = \frac{\mathrm{eV}^2\cdot\mathrm{m}}{\mathrm{eV}^2\cdot\mathrm{m}} = 1. \;\checkmark$$

Dimensionless, as required.

Why this had to work out

Inside sin, the argument *must* be dimensionless – otherwise the Taylor series $\sin x = x - x^3/6 + \cdots$ would mix incompatible units. The product $\hbar c$ exactly compensates the c^4 in $\Delta m^2 c^4$ when we use L in length units and E in energy units, leaving a pure number. This is why $\hbar c$ shows up in nearly every practical unit conversion in particle physics – it is the Rosetta stone between energy and length scales.

2.9 Step 2: Collect the numerical factor

From Eq. (1), separate numbers from variables:

$$\Phi = \underbrace{\frac{10^3}{4 \cdot 1.97327 \times 10^{-7} \cdot 10^9}}_{\equiv\, K} \cdot \frac{X \cdot Y}{Z}.$$

Simplify the numerical coefficient K:

$$K = \frac{10^3}{4 \cdot 1.97327 \times 10^2} = \frac{10^3}{789.308} = 1.2668\ldots$$

Since the problem asks for the result to two decimal places, $K \approx 1.27$.

Practical Form of the Oscillation Formula

$$\boxed{P(\nu_\alpha \to \nu_\beta) = \sin^2(2\theta)\,\sin^2\left[1.27\,\frac{\Delta m^2[\mathrm{eV}^2]\,L[\mathrm{km}]}{E[\mathrm{GeV}]}\right]}$$

The factor 1.27 arises from the combination $10^3/(4 \cdot \hbar c \cdot 10^9)$, where $\hbar c$ is in eV · m. It absorbs both (i) the 1/4 that came from the phase derivation and (ii) all the powers of 10 from converting eV → GeV and m → km.

Common Pitfall

You may see this formula written with both a 1/4 and a 1.27 in some textbooks (Thomson, problem 13.2 has a typo of this kind). That's incorrect double-counting. The 1.27 *already includes* the 1/4: don't divide by 4 again.

3 Part (b): MINOS Oscillation Fraction

3.1 Re-purposing the formula for $\nu_\mu \to \nu_\tau$

The two-flavor formula assumed mixing between states "1" and "2". We now reinterpret those as states "2" and "3" – physically describing the dominant atmospheric oscillation $\nu_\mu \leftrightarrow \nu_\tau$, controlled by the atmospheric mixing angle θ_{23} and the mass splitting $\Delta m^2_{23} \equiv m^2_3 - m^2_2$:

$$P(\nu_\mu \to \nu_\tau) = \sin^2(2\theta_{23})\,\sin^2\left[1.27\,\frac{\Delta m^2_{23}[\mathrm{eV}^2]\,L[\mathrm{km}]}{E_\nu[\mathrm{GeV}]}\right]. \tag{2}$$

3.2 Plug in the experimental parameters

Numerical inputs

- **Mixing:** $\sin^2(2\theta_{23}) = 1$ (maximal mixing, given).
- **Mass splitting (Griffiths Fig. 11.5, "Normal"):** $\Delta m^2_{23} \approx 2.5 \times 10^{-3}\,\mathrm{eV}^2$. (Griffiths' atmospheric value, Eq. 11.13. Newer global fits give 2.3–$2.5 \times 10^{-3}\,\mathrm{eV}^2$; the result is mildly sensitive to this choice – see Discussion.)
- **Baseline:** $L = 730\,\mathrm{km}$ (Fermilab to Soudan).
- **Beam energy:** $E_\nu = 3\,\mathrm{GeV}$ (constant, given).

3.3 Compute the argument Φ

$$\Phi = 1.27 \cdot \frac{\Delta m^2_{23} \cdot L}{E_\nu} = 1.27 \cdot \frac{(2.5 \times 10^{-3})\,(730)}{3}$$
$$= 1.27 \cdot \frac{1.825}{3} = 1.27 \cdot 0.6083$$
$$= 0.7726\ldots \quad \text{rad.}$$

This is the half-phase that goes inside the second sin. (For a sanity check, 0.7726 rad $\approx 44.3°$ – close to but not quite at the first oscillation maximum, which would require $\Phi = \pi/2 \approx 1.5708$.)

3.4 Compute the probability

$$\sin(\Phi) = \sin(0.7726) = 0.6979$$
$$\sin^2(\Phi) = (0.6979)^2 = 0.4870.$$

Therefore:

$$P(\nu_\mu \to \nu_\tau) = 1 \cdot 0.4870 = 0.487.$$

MINOS oscillation fraction

$P(\nu_\mu \to \nu_\tau) \approx 0.49 \iff \approx$ **49**% of ν_μ's have oscillated by the far detector.

Equivalently, the survival probability is $P(\nu_\mu \to \nu_\mu) \approx 1 - 0.49 = 0.51$ – about half the muon-neutrinos at 3 GeV make it through unchanged.

3.5 Interpretation

What MINOS Actually Measures

MINOS doesn't see ν_τ directly – producing a τ from ν_τ requires beam energies above the τ mass threshold (~ 3.5 GeV kinematically), and even at 3 GeV the cross section is very small. What MINOS measures is ν_μ *disappearance*: comparing the near-detector flux (before oscillation) to the far-detector flux. A $\sim 50\%$ deficit in the ν_μ rate at $E \sim 3$ GeV would be unmistakable evidence for oscillations – and indeed that is what was observed, fixing Δm^2_{23} to roughly the value we used.

3.6 Sensitivity to the Choice of Δm^2_{23}

The result depends fairly strongly on the assumed mass splitting. As a quick comparison:

Δm^2_{23} (eV^2)	Φ (rad)	$\sin^2\Phi$	$P(\nu_\mu \to \nu_\tau)$
2.3×10^{-3}	0.711	0.428	43%
2.4×10^{-3}	0.742	0.457	46%
2.5×10^{-3} (Griffiths)	**0.773**	**0.487**	**49%**

So the answer is in the 43% – 49% range depending on which textbook value you adopt; using the canonical Griffiths value gives $\sim 49\%$.

4 Summary

Two key takeaways

1. **Practical units form of Griffiths Eq. 11.9:**

$$P(\nu_\alpha \to \nu_\beta) = \sin^2(2\theta)\,\sin^2\left[1.27\,\frac{\Delta m^2[\mathrm{eV}^2]\,L[\mathrm{km}]}{E[\mathrm{GeV}]}\right].$$

   The factor $1.27 \approx 10^3/(4 \cdot 1.97 \times 10^2)$ *absorbs* both the 1/4 and the eV→GeV and m→km conversions.

2. **MINOS prediction:** With $L = 730\,\mathrm{km}$, $E = 3\,\mathrm{GeV}$, $\Delta m^2_{23} = 2.5 \times 10^{-3}\,\mathrm{eV}^2$, $\sin^2(2\theta_{23}) = 1$:

$$P(\nu_\mu \to \nu_\tau) \approx 0.49, \quad \text{i.e. } \sim 49\% \text{ of } \nu_\mu\text{'s oscillate before reaching the far detector.}$$